Cryptography

In cryptography, we protect data by converting it into an unreadable form.

Flow:

```
Plaintext → Encryption → Ciphertext → Decryption →
Plaintext
```

Plaintext

Original readable data

Can be:

- Text (e.g., "hello")
- Image (cat photo)
- Sensitive data (password, card info)

Ciphertext

Encrypted (scrambled) version of data
Not readable by humans

Example:

- x7#kP9!

Cipher

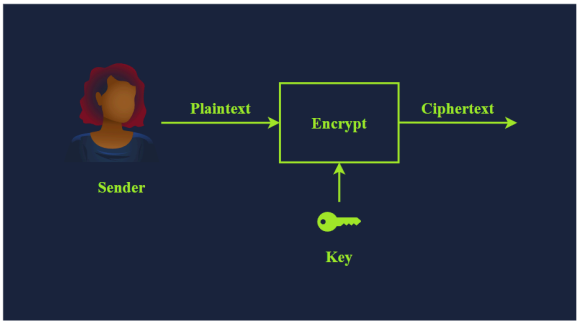
A mathematical algorithm used to:

- Encrypt data
- Decrypt data

Key

A secret value (bits/string) used in encryption & decryption

- Cipher → public
- Key → must be secret



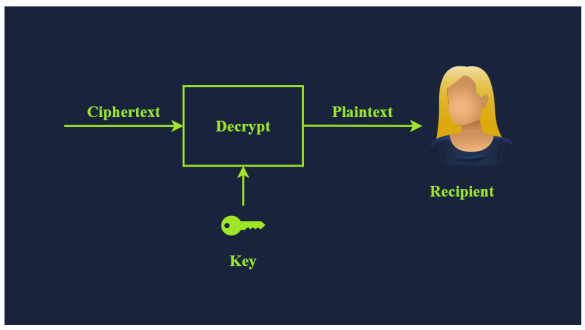
Encryption

Process of converting:

- Plaintext → Ciphertext

Using:

- Cipher + Key



Decryption

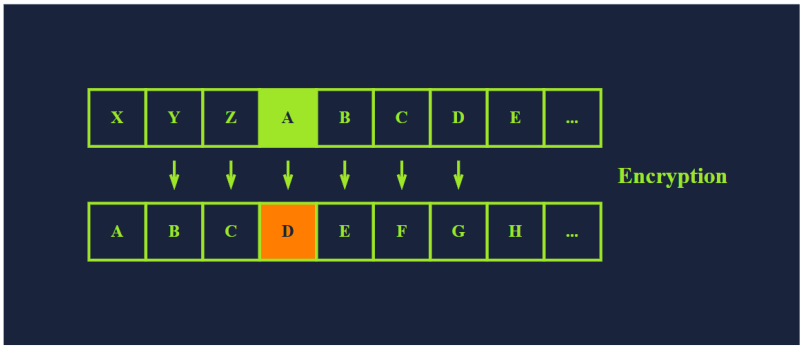
Reverse process:

- Ciphertext → Plaintext

Using:
- Same (or related) key

Caesar Cipher

Caesar Cipher is a simple encryption technique where each letter is shifted by a fixed number (key) in the alphabet.



Example

Plaintext: `TRYHACKME`

Key: 3 (right shift)

Encryption:

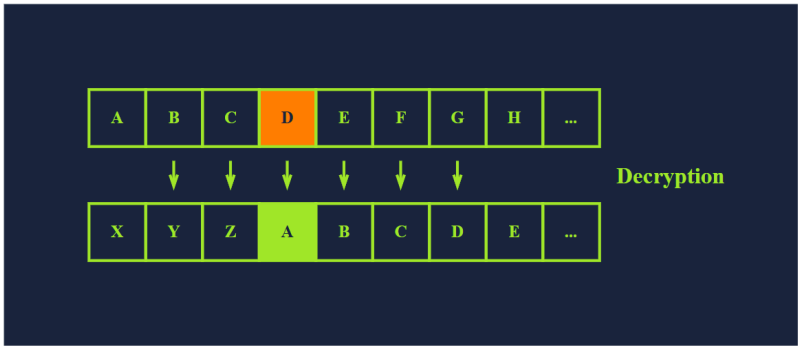
```
T → W
R → U
Y → B
```

Ciphertext:

```
WUBKDFNPH
```

Encryption & Decryption

- Encryption → shift letters right
- Decryption → shift letters left

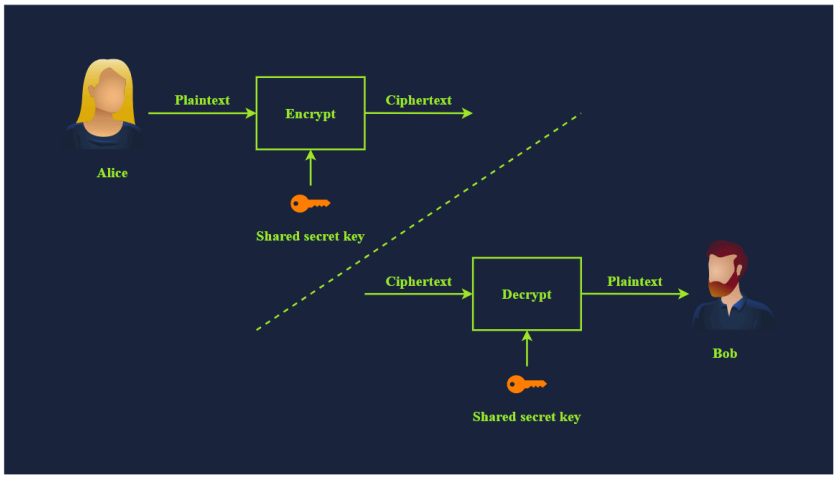


Symmetric vs Asymmetric Encryption

Symmetric Encryption (Private Key)

Uses same key for:

- Encryption
- Decryption



Plaintext + Key → Ciphertext
Ciphertext + Same Key → Plaintext

Common Algorithms

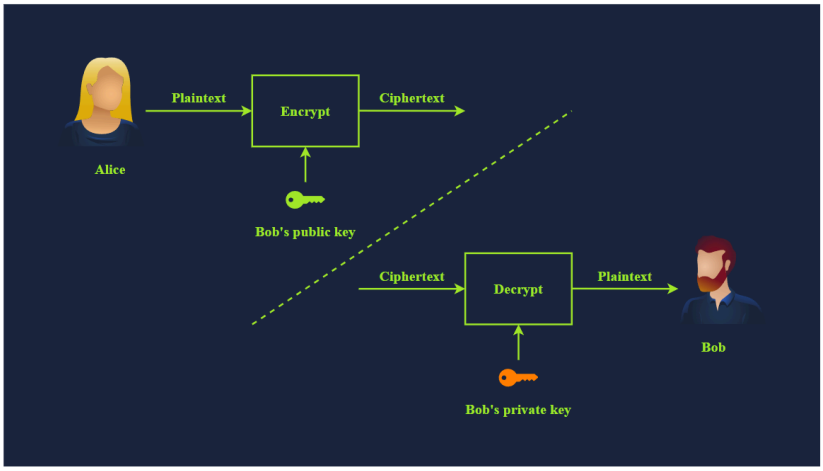
- DES
- 3DES
- AES (most secure & modern)

DES → 56-bit (broken, insecure)
3DES → stronger but outdated
AES → 128 / 192 / 256-bit (recommended)

Asymmetric Encryption (Public Key)

Uses two different keys:

- Public Key → Encryption
- Private Key → Decryption



Plaintext + Public Key → Ciphertext
Ciphertext + Private Key → Plaintext

Common Algorithms

- RSA
- Diffie-Hellman
- ECC (Elliptic Curve Cryptography)

Key Size Info

- RSA → 2048+ bits
- Diffie-Hellman → 2048+ bits
- ECC → smaller keys, same security